STAGE B1 — MINI PROJECT B

SMRTTECH 4SC3 Smart Cities and Communities

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This document contains all the evidence required to prepare Stage B1 as part of the Mini Project B. Decisions that impact the development of Stage B1 are outlined and agreed upon by all group members. All work was done by the members of Group 2, only using AI when required by the instructions.

Date: 11/05/2026

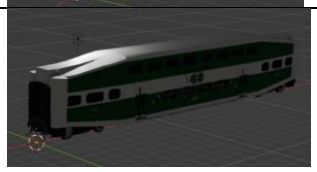
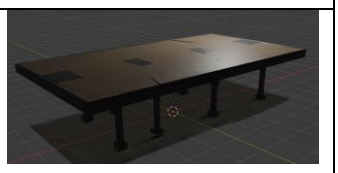
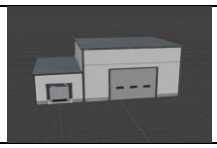
Concept

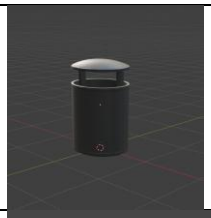
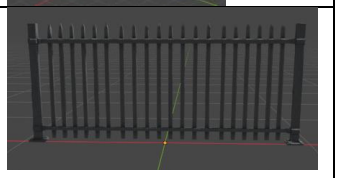
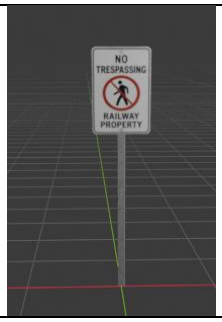
For this project, our group has selected to model a passenger locomotive—specifically a GO Train, as these frequent the Oshawa GO station.

The planned location of our generated assets will be as follows:



Asset Inventory

Asset ID	Asset Name	Responsible Member	Dimensions (in m)	Blender/FBX Path	Method/Source	Preview Image
A01	Locomotive	Nicolas	20.73L x 4.8W x 6.43H	.\Group02_StageB1\02_Exports\A01_Locomotive	Meshy AI + Blender Formatting (from Google Images)	
A02	Coach car	Nicolas	25.9L x 3.39W x 5.0H	.\Group02_StageB1\02_Exports\A02_CoachCar	Meshy AI + Blender Formatting (from Google Images)	
A03	Station building	Nicolas	55.44L x 36.0W x 14.3H	.\Group02_StageB1\02_Exports\A03_StationBuilding	Meshy AI + Blender Formatting (from Google Images)	
A04	Station platform	Nicolas	9.05L x 4.88W x 2.17H	.\Group02_StageB1\02_Exports\A04_StationPlatform	Meshy AI + Blender Formatting (from Google Images)	
A05	Vegetation - Tree	Tommy	1.1L x 1.08W x 1.89H	.\Group02_StageB1\02_Exports\A05_Tree	Meshy AI + Blender Formatting (from ChatGPT)	
A06	Vegetation - Shrub	Tommy	1.75L x 1.78W x 0.741H	.\Group02_StageB1\02_Exports\A06_Shrub	Meshy AI + Blender Formatting (from ChatGPT)	
A07	Surrounding buildings - commercial	Tommy	16L x 10W x 7H	.\Group02_StageB1\02_Exports\A07_CommercialBuilding	Meshy AI + Blender Formatting (from ChatGPT)	
A08	Surrounding buildings - industrial	Tommy	25L x 15W x 9H	.\Group02_StageB1\02_Exports\A08_IndustrialBuilding	Meshy AI + Blender Formatting	

					(from ChatGPT)	
A09	Street furniture - bench	Tommy	2.5L x 0.8W x 1.5H	.\Group02_StageB1\02_Exports\A09_Bench	Meshy AI + Blender Formatting (from ChatGPT)	
A10	Street furniture – Waste Bin	Tommy	1L x 0.9W x 1.5H	.\Group02_StageB1\02_Exports\A10_WasteBin	Meshy AI + Blender Formatting (from ChatGPT)	
A11	Lighting object	Jacob	2.34L x 2.34W x 15H	.\Group02_StageB1\02_Exports\A11_LightingObject	Meshy AI + Blender Formatting (from Copilot)	
A12	Railway infrastructure	Jacob	1.9L x 0.113W 0.806H	.\Group02_StageB1\02_Exports\A12_RailwayFence	Meshy AI + Blender Formatting (from Copilot)	
A13	Railway warning sign	Jacob	0.575L x 0.0485W x 2.41H	.\Group02_StageB1\02_Exports\A13_WarningSign	Meshy AI + Blender Formatting (from Copilot)	

Milestone 1

A01 Prompt

No prompt was used to generate the 2D image as we did not use generative AI to create the 2D image.

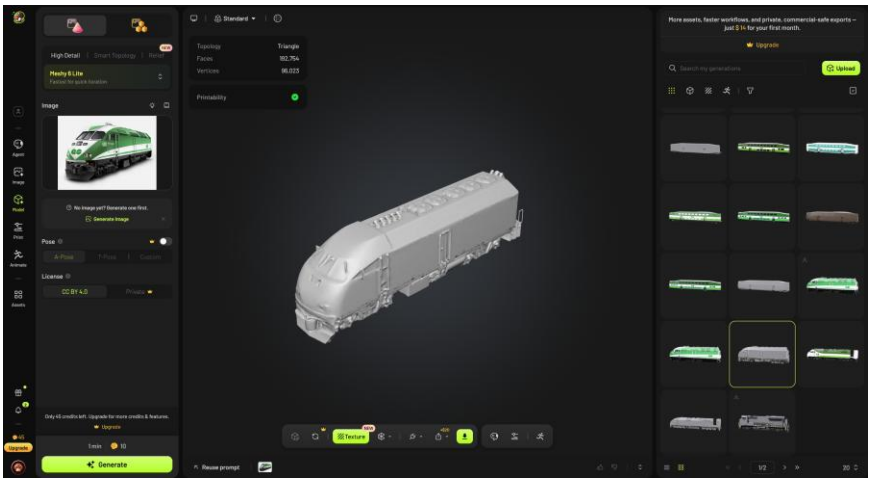
Instead, the source for the reference image was taken from Google Images, from this specific site: https://ignitiondigital.com/portfolio/go-transit-locomotive

Reference Image

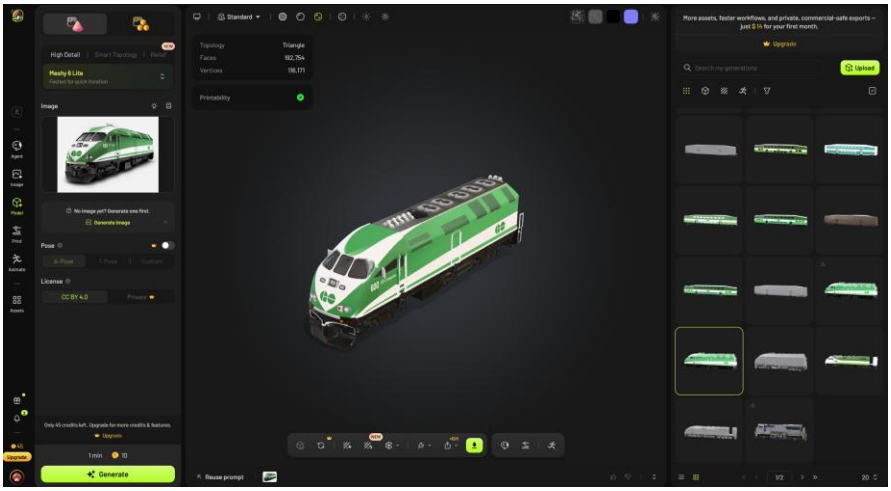


This is the reference image we used to create the locomotive (A01). We fed this into Meshy AI and it generated the 3D model. We used the same reference image as an image input for the texture of the model.

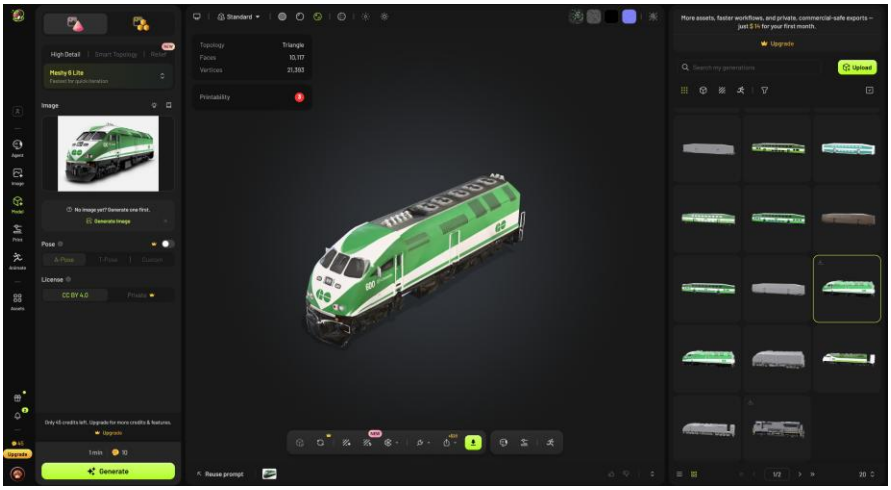
Generated/Textured Previews



This is the 3D model Meshy AI generated with our 2D reference image.



This is the 3D model with the texture now added, which we created through an image input of the same 2D reference image.



This is the final re-meshed 3D model we downloaded. As you can see, the number of faces and vertices have drastically decreased. This was done to optimize the models, as a greater number of faces and vertices on each object can slow down the game engine.

Before-and-After Mesh Statistics

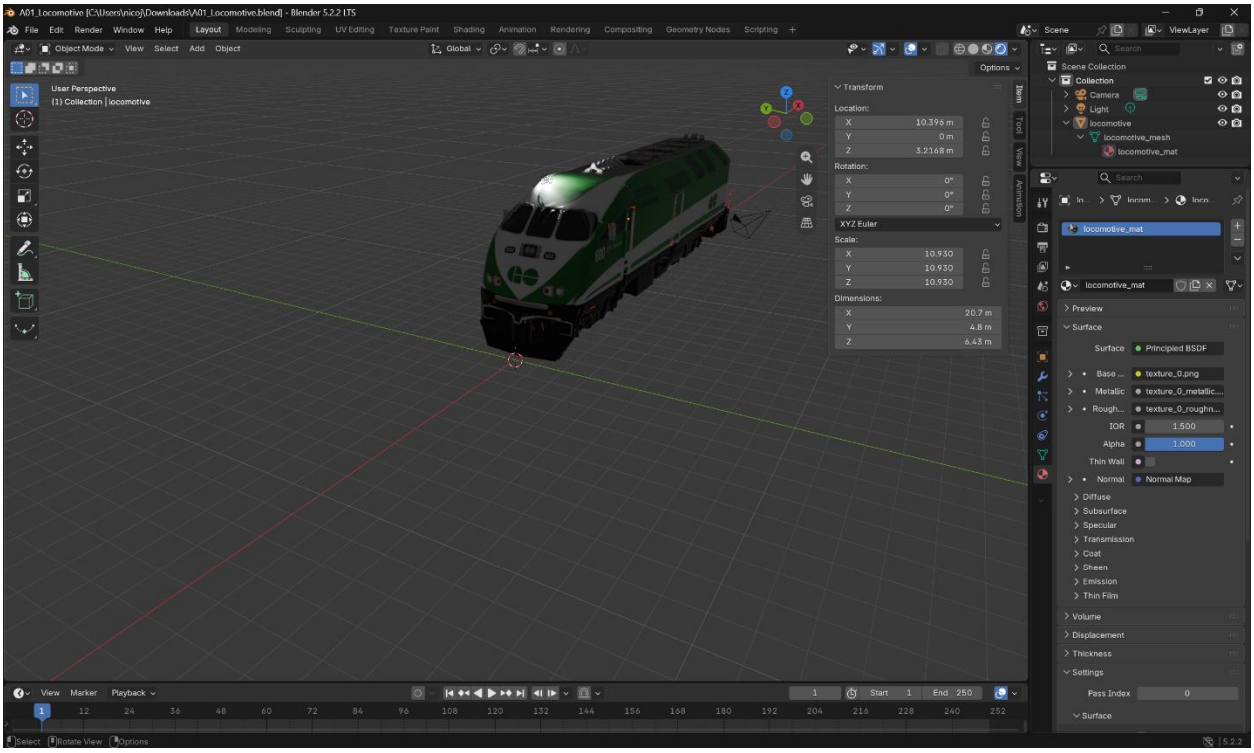
Version	**Faces**	**Vertices**	**Notes**
Original (Meshy)	192,754	116,171	Full detail
Reduced	10,117	21,393	Detail holds surprisingly well. Text is slightly blurrier. Overall texture still looks clean.

We chose this first generated model of the locomotive from Meshy AI because it did it surprisingly well on the first try (unlike A02, the coach car, which took many tries to get a good model and texture). The text looks clean and is readable, the colours are vibrant, and the shape of the locomotive is as desired.

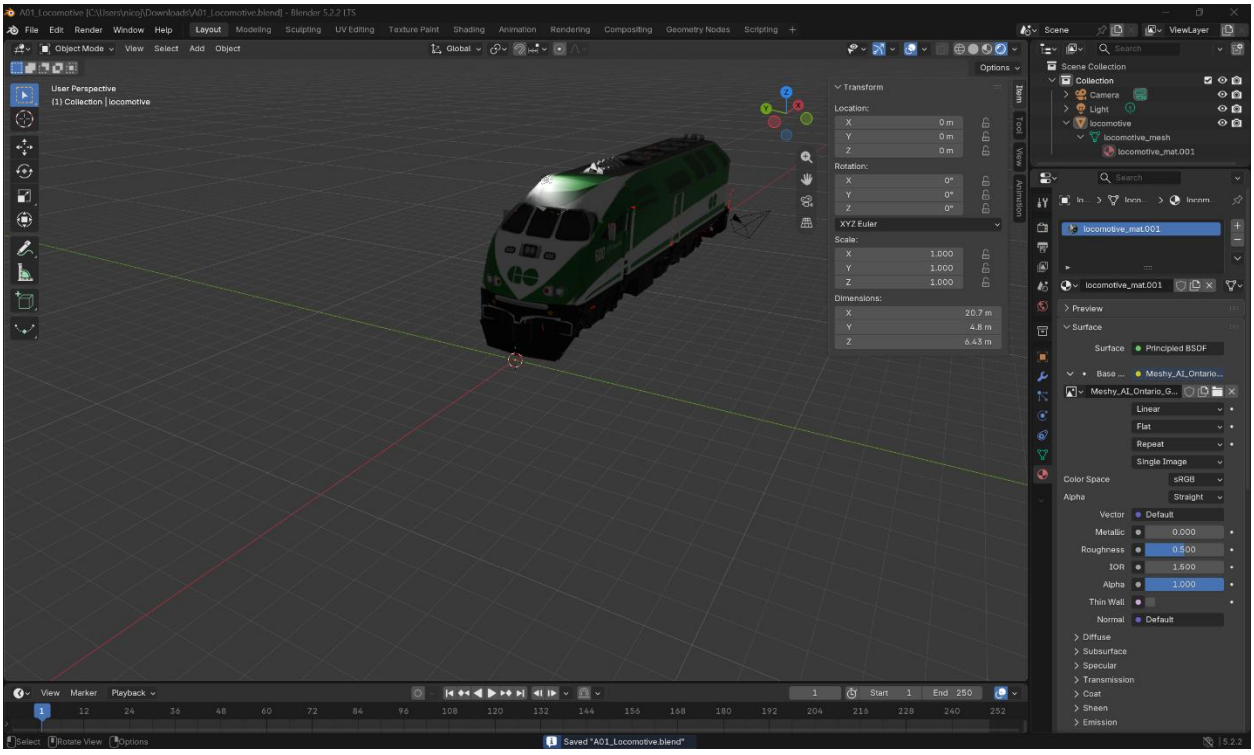
There were no changes to external models because the 2D reference image we chose, our only external model input for Meshy AI, produced a reasonably well-made model and texture on the first try.

Milestone 2

Before-and-After of Scale/Alignment

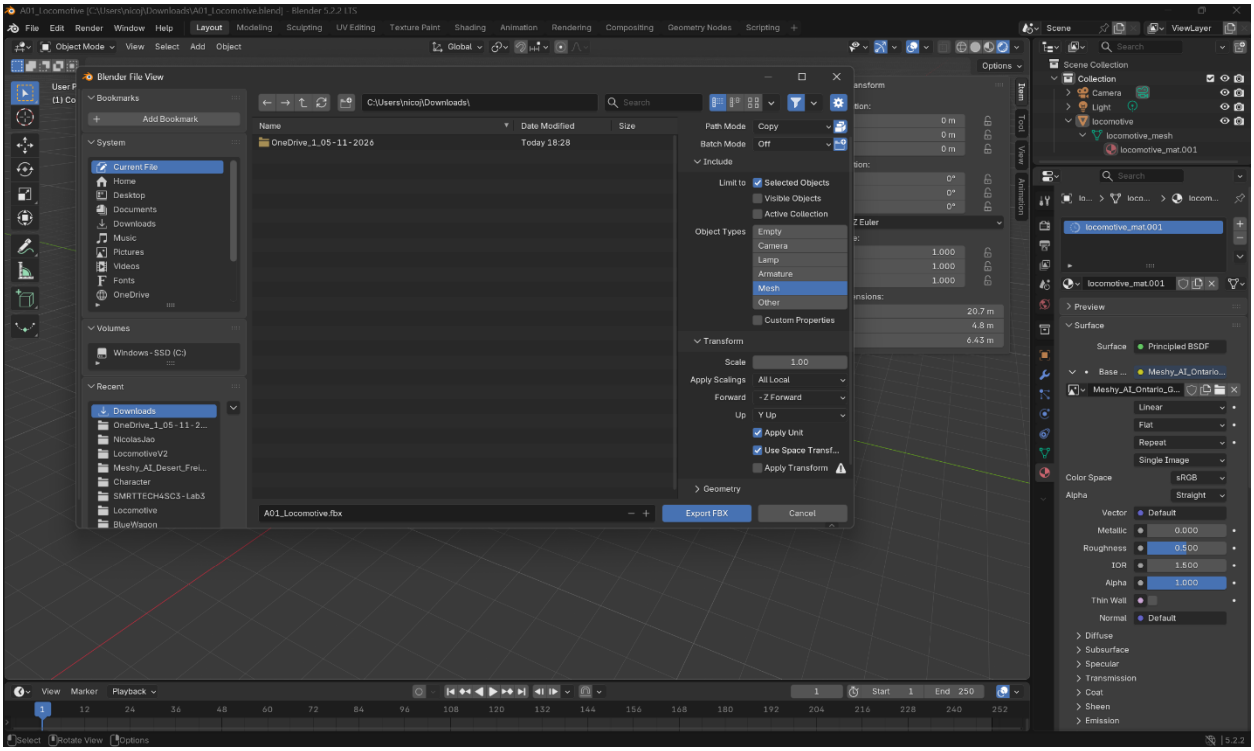


This is the fbx model imported into blender before the scaling transforms have been applied.



This is the model after scaling it properly and applying all transforms. As you can see, we also renamed the files properly and created a new material to add the texture. It is also aligned on all axes.

Export Settings



These are the export settings we used when exporting the final model as an fbx.

This process was repeated for all assets.

Unresolved Issues

There were no unresolved issues. There were optional features we chose not to apply, and some features where we did not have to do anything, but we understand (UV mapping).